

SPECIFICATION FOR PROCUREMENT OF 800 GALLON PURE WATER TANK

1. Scope. Provide an eight-hundred (800) Gallon Stainless Steel low pressure horizontally oriented cylindrical tank with elliptical end caps. Tank assembly will include tank saddle with integral lift lugs (tank & saddle) and fork lift tubes within the saddle base. Tank assembly shall be outfitted with continuous level & temperature indication, along with service connections detailed within this statement of work (inlet, outlet, vent, sample, nitrogen supply, etc.). Overall tank height and dimensions shall allow placement within a standard width and height conex box/sealand container.
2. References.
 - 2.1 ASME Boiler and Pressure Vessel Code, Section VIII Division 1
 - 2.2 MIL-STD-2041, Control of Detrimental Materials
 - 2.3 MIL-STD-767, Control of Hardware Cleanliness
 - 2.4 ASME Boiler and Pressure Vessel Code, Section II Part A (Material Specification SA-240)
 - 2.5 ASTM A500, Standard Specification for Cold-formed Welded and Seamless Carbon Steel Structural Tubing in Rounds and Shapes
 - 2.6 ASTM A36, Standard Specification for Carbon Structural Steel
 - 2.7 ASTM A240/A240M, Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels
 - 2.8 ASTM A276/A276M, Standard Specification for Stainless Steel Bars and Shapes
 - 2.9 ASTM A511/A511M, Standard Specification for Seamless Stainless Steel Mechanical Tubing and Hollow Bar
 - 2.10 MIL-PRF-23236, Coating Systems for Ship Structures
 - 2.11 A-A-59326D, Coupling Halves, Quick-Disconnect, Dam-Locking Type, General Specification For
 - 2.12 ASME B16.5 Pipe Flanges and Flange Fittings
 - 2.13 ASTM A312, Standard Specification for Seamless, Welded, and Heavily Cold Worked Austenitic Stainless Steel Pipes
 - 2.14 ASME B16.9 Factory-Made Wrought Butt welding Fittings
 - 2.15 ANSI N14.6-1993, Special Lifting Devices for Shipping Containers Weighing 10000 Pounds (4500 kg) or More
 - 2.16 NAVSEA Technical Publication T9074-AS-GIB-010/271, Requirements for nondestructive Testing Methods
 - 2.17 MIL-STD-2035A, Non-Destructive Testing Acceptance Criteria.

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- 2.18 ASTM A967/A967M, Standard Specification for Chemical Passivation Treatments for Stainless Steel Parts
- 2.19 SAE J514, Hydraulic Tube Fittings
- 2.20 AWS D1.1, Structural Welding Code – Steel
- 2.21 AWS D1.2, Structural Welding Code – Aluminum
- 2.22 AWS D1.6, Structural Welding Code – Stainless Steel
- 2.23 MIL-STD-22, Welded Joint Design

3. General Requirements.

- 3.1 Where specifications and/or standards are specified or referenced, the latest revision shall be used unless otherwise specified.
- 3.2 Tank design, fabrication and testing shall be in accordance with reference 2.1. Tank shall be stamped with the code “U” symbol.
- 3.3 Formal NNSY approval, via signature, of tank assembly drawings, with material list(s), shall be obtained prior to beginning fabrication.
- 3.4 All internal and/or fluid contacting surfaces/materials of the tank assembly (tank body, end caps, piping, etc.) and its components (e.g. valves, level/temperature transmitter assembly, etc.) shall meet detrimental material and cleanliness requirements of references 2.2 and 2.3.
 - 3.4.1 Interior tank and piping surfaces (i.e., fluid contacting surfaces) shall be cleaned to remove debris, foreign materials, markings, corrosion, weld spatter, paint, preservatives, oils, greases, lubricates, etc. in accordance with requirements of reference 2.3.
 - 3.4.2 All lubricants for component operation (valve, fittings, etc.) or shop assembly shall meet detrimental material requirements of reference 2.2 or be completely removed per 3.4.1.
 - 3.4.3 All Lubricants used for moving parts within valves/fittings/components shall be formally authorized and agreed to, via signature, between NNSY and the vendor.
- 3.5 All equipment/components furnished shall NOT contain mercury. In addition, equipment/components furnished shall NOT have come into direct contact with mercury or mercury containing components during fabrication or testing.
- 3.6 Tank shell/body and elliptical end caps shall be stainless steel, Grade 316 (Reference 2.4 or equivalent).
- 3.7 Tank assembly saddle, lift lugs, base with fork lift tubes and framing for process meter enclosure mounting shall be fabricated from carbon steel (references 2.5, 2.6 or equivalent) or grade 304/316 stainless steel (references 2.7, 2.8, 2.9 or equivalent).
- 3.8 Coating systems (i.e., paints and primers) used on carbon steel structural/framing components shall be non-hazardous per Resource Conservation and Recovery Act (RCRA) requirements.

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- 3.8.1 Norfolk Naval Shipyard (NNSY) shall approve coating system, via signature, prior to coating system use/application. Coating systems per reference 2.10 are approved for use.
- 3.8.2 Prohibited colors are yellow, magenta, purple, gold, red and orange.
- 3.9 Markings (e.g., weight, lift points, serial numbers, etc.), when referenced to this paragraph, shall be made using 1 inch tall letters and highly contrasting color paint (e.g., black frame with white lettering). Label plates/placards made using Gravotech Gravoply, or equivalent poly/composite material providing high contrast (e.g., black background with white lettering) may be used for marking on stainless steel structures and components.
- 3.10 The use of cadmium plated components/fittings is prohibited.
- 3.11 The use of brass/bronze components or fittings is prohibited. Any use of brass/bronze components or fittings shall be formally authorized and agreed to, via signature, between NNSY and the vendor.
- 3.12 Teflon tape or any type of thread sealant/locker shall NOT be used on any fluid contacting components (e.g., NPT Joints, etc.) unless formally authorized and agreed to, via signature, between NNSY and the vendor. Loctite 5772 is an authorized thread sealant/locker which may be used on fluid contacting components.
- 3.13 All Camlock fittings shall be stainless steel and meet requirements of reference 2.11. Rings and pins on camlock arms shall NOT be cadmium plated.
- 3.14 All Flanges shall be stainless steel (grade 316), raised face with weld neck ends for connection to piping per reference 2.12.
- 3.15 All piping assembly end fittings (open end of assembly for connecting temporary equipment/systems) shall be provided with a mechanical cap/blank installed (i.e., SAE J514 flare cap, blind flange, union blank) as applicable.
- 3.15.1 All Camlock closure dust caps shall be Dixon Cam and Groove EZ-Link armless dust caps. Caps shall be installed on all pipe assembly ends containing male Camlock adapters.
- 3.15.2 All mechanical caps/blanks shall be attached to their respective piping assembly via lanyard (stainless steel braided wire (1/16 – 1/8 inch diameter) and nicopress sleeving). Flange blanks are not required to be attached via lanyard.
- 3.16 All piping shall be seamless stainless steel (grade 316) in accordance with reference 2.13. Minimum pipe schedule allowed is SCH 40.
- 3.17 All pipe fittings shall be stainless steel (grade 316) butt weld end prep per reference 2.14, unless otherwise specified in this statement of work or authorized by NNSY.
- 3.18 Valves shall be three piece stainless steel (grade 316) full port ball valves with butt weld ends. Seats, seals and packing shall be PTFE. All valves shall be Swagelok 60 series or Bray 7000/8000 series, with locking lever handles, meeting component design pressures.
- 3.19 Tank assembly design/operating pressures, flowrates and temperature requirements are as specified below.

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- 3.19.1 Design pressure: 30 psig.
- 3.19.2 Vacuum pressure, design: 29 inches Hg/full vacuum.
- 3.19.3 Normal operating pressure: 20 psig maximum.
- 3.19.4 Flowrate via low point suction connections: 60 GPM.
- 3.19.5 Design temperature is -20° Fahrenheit to 120° Fahrenheit ambient outside air temperature. Fluid will be at ambient temperature (e.g., typically 50° Fahrenheit or greater with freeze protection).
- 3.20 Overall tank dimensions shall allow tank to be located within a standard conex/sealand container. Dimensions on one end of the tank assembly (height and width) shall allow it to pass through a standard conex/sealand container door opening (width – 92 inches; height – 89 inches).
- 3.21 Use of the following during preparation and fabrication of the tank is prohibited.
 - 3.21.1 Oil or other cutting fluids.
 - 3.21.2 Metal removal or finishing tools made of materials which may cross contaminate surfaces or cause surface corrosion staining of stainless steel surface, when exposed to water or a moist atmosphere.
 - 3.21.3 Power driven tools or wire brushes that tend to smear metal.
 - 3.21.4 Tools or abrasive materials which have previously been used on low alloy steels, aluminum, lead or other low melting point materials.
- 3.22 All testing requiring liquid supplied to the tank and piping internals shall be performed using deionized water.
- 3.23 Vendor shall provide NNSY representative(s) the opportunity to perform an inspection of the accessible internal and external surfaces of the tank assembly. NNSY reserves the right to witness fabrication and testing of the tank assembly.

4. Tank Assembly Base Plate, Framing and Saddle.

- 4.1 Tank assembly, saddle and any support component mounting frames (i.e., process meter enclosure, power box) shall be built on a main assembly base plate/frame.
- 4.2 Base plate/frame shall be constructed from plate/shaped structural carbon steel (references 2.5, 2.6 or equivalent) of grade 304/316 stainless steel (references 2.7, 2.8, 2.9 or equivalent).
- 4.3 Base plate/frame/lift lugs shall be coated to prevent corrosion per requirements of paragraph 3.8. This requirement is not applicable to stainless steel plate/frame/lift lug materials.
- 4.4 Base plate/frame shall be outfitted with fork lift tubes which run in both directions (length and width of base) to allow movement of the assembly via forklift from all sides.

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- 4.5 Base plate/frame shall be sized to ensure the tank assembly (all components, piping, valves and saddle with tank at 90% fill volume) shall not create a floor/ground loading which exceeds 400 lbs/ft². Base plate/frame shall be sized to transit through a standard conex/sealand container door opening width of 92 inches on one end.
- 4.6 All exterior surfaces of base plate, framing and saddle shall be smooth and all sharp edges rounded/chamfered.

5. Tank Assembly Lift Lugs/Rigging Attachment Points and Non Destructive Test Details.

- 5.1 Tank assembly shall be outfitted with lift lugs/rigging attachment points to allow entire assembly (tank and base plate/frame) to be lifted via an overhead crane.
 - 5.1.1 Lift lugs/rigging attachment points shall be permanently attached to the base plate/frame or the tank saddle support members.
 - 5.1.2 Lift lugs/rigging attachment points shall be oriented to allow use of a single overhead crane attachment point or a spreader beam.
- 5.2 Integral lift lugs/rigging attachment points shall meet the requirements of reference 2.15.
- 5.3 Vendor shall provide certification and results of the drop weight test per ASTM E208 or a Charpy Impact test per ASTM A370. The anticipated minimum service temperature for the tank assembly and integral lift lugs/rigging attachment points is 20° Fahrenheit.
- 5.4 Lift lugs/rigging attachment points and load-bearing members shall be capable of supporting a load three times the combined weight of the empty tank assembly (3:1 yield), without generating a combined shear stress or maximum tensile stress at any point in the device exceeding the corresponding minimum tensile yield strength of the materials of construction.
- 5.5 Lift lugs/rigging attachment points and load-bearing members shall also be capable of lifting five times the weight of the empty tank assembly (5:1 ultimate) without exceeding the ultimate tensile strength of the materials of construction.
- 5.6 Load Testing and Non Destructive Testing (NDT) of the lift lugs/rigging attachment points.
 - 5.6.1 Vendor shall perform pre and post load test visual inspection of the lift lugs/rigging attachment points; no signs of cracked, broken, worn, distorted, corroded, deteriorated, binding or loose parts are allowed.
 - 5.6.2 Vendor shall perform pre and post load test magnetic particle or liquid penetrant (stainless steel materials) inspections of the entire lift lugs/rigging attachment points and attachment welds. Magnetic particle or liquid penetrant inspections shall be performed in accordance with reference 2.16 with acceptance criteria per reference 2.17 (Class 3 or equivalent for welds). Certified report required.

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- 5.6.3 Vendor shall perform a load test of the integral lift lugs/rigging attachment points. Test load shall be performed at 150% (+5/-0%) of the maximum service load. Maximum service load for this application is the weight of the empty tank assembly. Load test shall be held for a minimum of 10 minutes. Certified report required.
- 5.7 Vendor shall provide a lifting/rigging diagram detailing the preferred configuration (sling lengths, angles, minimum pendent/component capacity, etc.).

6. Tank Details.

- 6.1 Tank shall be an eight hundred (800) gallon nominal capacity horizontal cylindrical tank with elliptical end caps.
- 6.2 Tank shell, including elliptical end caps shall be stainless steel (grade 304/316) per reference 2.4, or an equivalent stainless steel plate specification. Minimum acceptable plate thickness for the shell and elliptical end caps is 5/16 inch. Plate with a No. 4 or finer finish, for use on the inside tank surface, shall be used to facilitate obtaining the required interior surface roughness/finish of paragraph 6.3.
- 6.3 Interior tank surface (including piping and components) shall have a surface roughness/finish of 32 Ra or better. All pits, gouges, scratches, tool marks, weld spatter, etc., shall be faired into the surrounding surface.
- 6.4 Interior tank weld surface(s) (including piping and components) shall have a roughness/finish of 32 Ra or better. Welds shall be free of undercut and shall not snag and/or tear a cleaning cloth when wiped in any direction.
- 6.5 A GAR S-22 microfinish comparator surface roughness scale (GAR Electroforming Division Danbur, CT WWW.garelectroforming.com) or similar comparator shall be used to verify interior tank and weld surface finish.
- 6.6 Tank shall have a personnel access for tank entry which is a minimum of 18 inches in diameter. Closure seal shall utilize a Buna-N gasket. Use of cork-rubber gaskets is prohibited.
- 6.7 Interior and exterior of the tank shell and elliptical end caps shall be free of embedded ferrous or aluminum particles resulting from rolling and/or forming. Rollers and/or forming equipment used exclusively on stainless steel should be used to form the tank.
 - 6.7.1 Rollers or other forming machines that have been used on ferrous or aluminum metals may be used to form the tank shell and end caps provided special precautions are taken to ensure a satisfactory barrier is located between the rollers/forming machines and the tank shell or elliptical end cap materials.
 - 6.7.2 The contractor shall submit a written request to NNSY for use of rollers or other forming machines discussed in paragraph 6.7.1 and obtain/receive written approval from NNSY. Request shall specify why use of the paragraph 6.7.1 equipment is required and provide details of the special precautions which will be in place ensuring contact between the tank shell/elliptical end cap material and the rollers/forming equipment does not occur.

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- 6.8 Tank shell and elliptical end cap base materials and weld areas (prior to and after completion of welding) shall be passivated per reference 2.18, or other equivalent specification. Solutions and passivation process shall maintain compliance with requirements of references 2.2 and 2.3. Certified report required.

7. Piping Assemblies and Connection Details.

- 7.1 Tank shall contain two (2) drain/suction piping assemblies connected to tank via bottom connections/penetrations and terminate with end caps on the same tank end.
- 7.1.1 Piping assemblies shall be 1 ½ inch SCH 40 per paragraph 3.16.
- 7.1.2 Piping assemblies shall have a 1 ½ inch full port ball valve, per paragraph 3.18, between tank penetration/connection and end fitting.
- 7.1.3 Process connection end fitting shall be a 1 ½ inch male camlock per paragraph 3.13. Fitting shall be provided with an end cap closure per paragraph 3.15.
- 7.1.4 Each penetration (inlet to discharge piping assembly inside tank) shall have a vortex break installed.
- 7.1.5 One piping assembly tank penetration shall be on longitudinal centerline of tank bottom and the other shall be slightly off centerline and parallel with the other assembly (see figure 1). See figure 2 for general piping assembly arrangement.
- 7.2 Tank shall contain a single piping assembly for obtaining liquid samples, connected to tank via bottom connection/penetration and terminate with end cap at same tank end as paragraph 7.1 piping assemblies (see figure 1). See figure 2 for general piping assembly arrangement.
- 7.2.1 Piping assembly shall be 3/8 inch SCH 40 per paragraph 3.16.
- 7.2.2 Piping assembly shall have a 3/8 inch full port ball valve, per paragraph 3.18, between tank penetration/connection and end fitting.
- 7.2.3 Process connection end fitting shall be a 3/8 inch 37 degree flare fitting per reference 2.19 (i.e., SSP PN BM6-6J or equivalent). Fitting shall be provided with an end cap closure per paragraph 3.15 (i.e., SSP PN AS6Z-2 or equivalent).
- 7.2.4 Tank penetration shall be slightly off centerline of tank bottom longitudinal; piping assembly shall also parallel assemblies of paragraph 7.1 (see figure 1). See figure 2 for general piping assembly arrangement.

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Figure (1): Drain/Suction and Sample Line Arrangement; Looking from Tank End

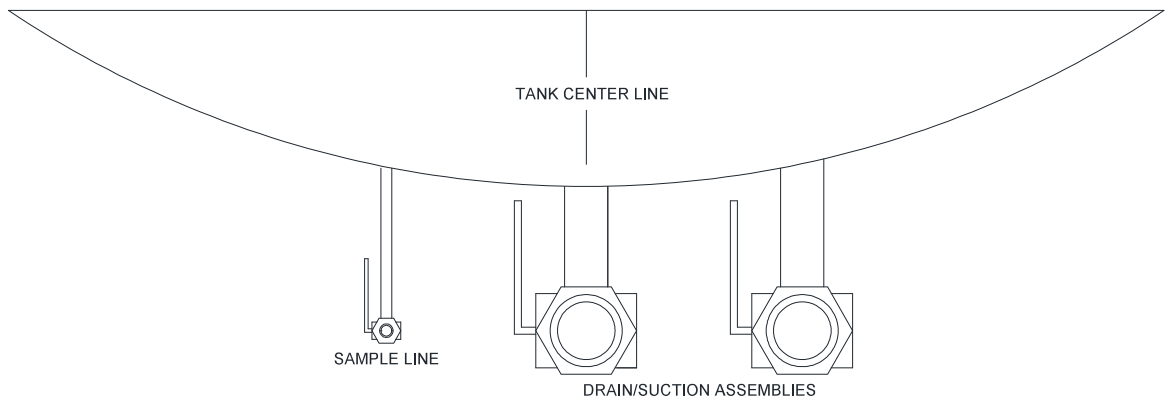
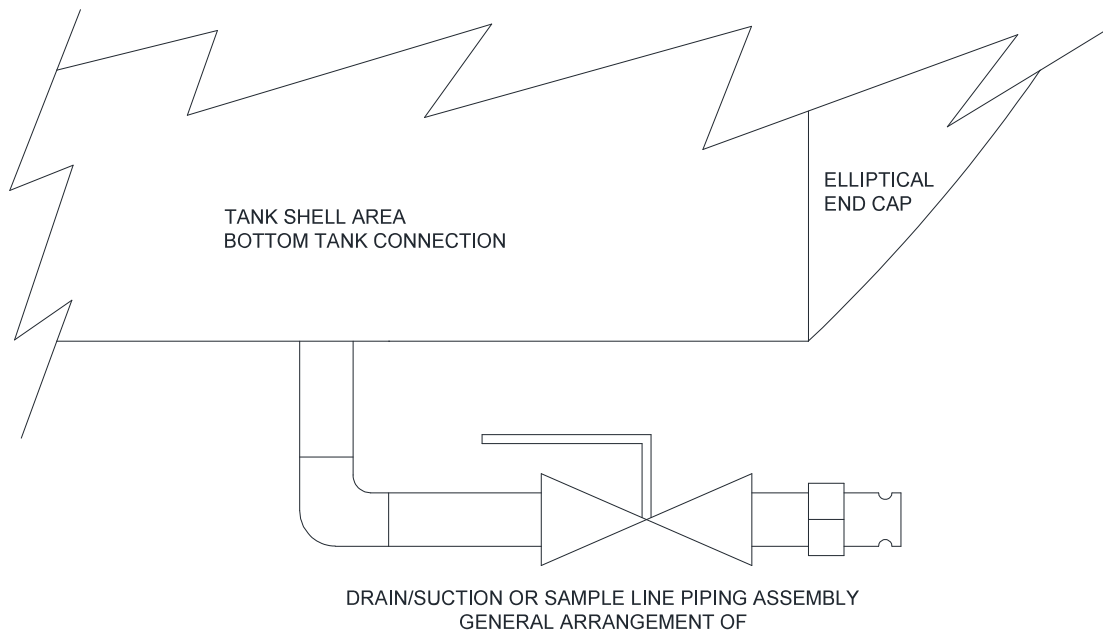


Figure 2: General Piping Assembly Arrangement for Drain/Suction and Sample Line



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- 7.3 Tank shall contain a single fill line assembly connected to tank via top connection/penetration and terminate with end cap on the same tank end (See figure 3). See figure 4 for general piping assembly arrangement
 - 7.3.1 Fill assembly shall be located at the tank end opposite the piping assemblies of paragraphs 7.1 and 7.2.
 - 7.3.2 Piping assembly shall be 2 inch SCH 40 per paragraph 3.16.
 - 7.3.3 Piping assembly shall have a 2 inch full port ball valve, per paragraph 3.18, between tank penetration/connection and end fitting.
 - 7.3.4 Process connection end fitting shall be a 2 inch male camlock x butt weld, per paragraph 3.13. Piping assembly shall be supplied with an end cap closure fitting per paragraph 3.15.
 - 7.3.5 Piping assembly shall maintain a constant downward slope from piping assembly camlock connection to tank top penetration.
 - 7.3.6 Piping assembly shall be slightly off centerline and parallel with the piping assembly of paragraph 7.4. See figures 3 and 4 for piping assembly arrangement details.
- 7.4 Tank shall contain a single piping assembly for discharge receipt from a relief valve.; connected to tank via top connection/penetration and terminate with end cap at same tank end as the paragraph 7.3 piping assembly (see figure 3). See figure 4 for general piping assembly arrangement.
 - 7.4.1 Piping shall be 1 inch SCH 40 per paragraph 3.16.
 - 7.4.2 Piping assembly shall have a 1 inch full port ball valve, per paragraph 3.18, between tank penetration/connection and end fitting.
 - 7.4.3 Process connection end fitting shall be a 1 inch 37 degree flare fitting per reference 2.19 (i.e., SSP PN BM16-16J or equivalent). Fitting shall be provided with an end cap closure per paragraph 3.15 (i.e., SS PN AS16Z-2 or equivalent).
 - 7.4.4 Piping assembly shall maintain a constant downward slope from piping assembly 37 degree flare connection to tank top penetration.
 - 7.4.5 Piping assembly shall be slightly off centerline and parallel with piping assembly of paragraph 7.3. See figures 3 and 4 for piping assembly arrangement details.

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Figure 3: Fill and Relief Valve Discharge Receipt Lines; Looking from Tank End.

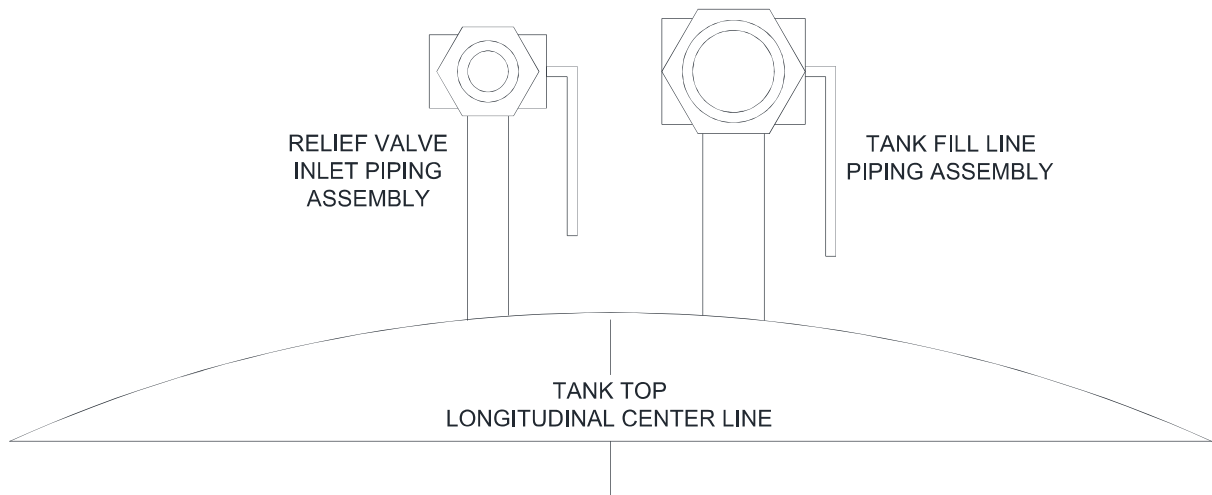
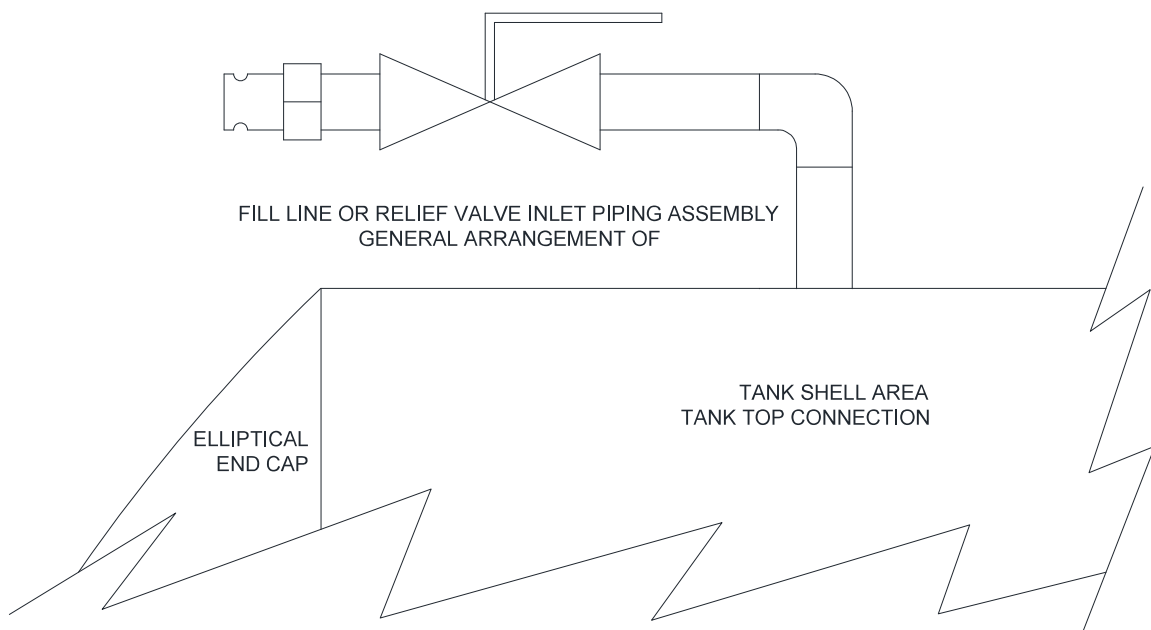


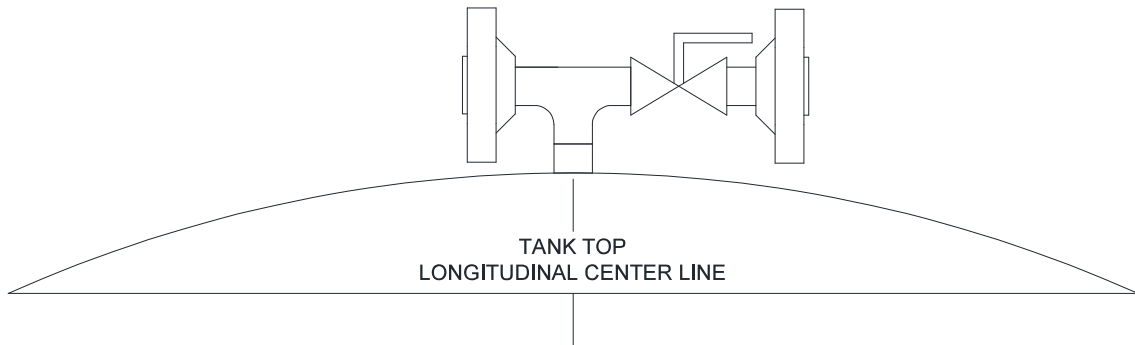
Figure 4: General Piping Assembly Arrangement for Fill and Relief Valve Discharge Receipt Lines.



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- 7.5 Tank shall contain a single piping assembly connected via tank top connection/penetration for a nitrogen supply and installation/connection of a make-up air assembly. See figure 5 for piping assembly arrangement.
- 7.5.1 Piping for nitrogen supply shall be 1 inch SCH 40, per paragraph 3.16.
- 7.5.2 Piping for nitrogen supply shall have a 1 inch full port ball valve per paragraph 3.18, between tank penetration/connection and end fitting.
- 7.5.3 Piping for make up air assembly shall be 1 inch SCH 40, minimum, per paragraph 3.16. Piping shall be sized to ensure that negative pressure/vacuum condition is not created during liquid removal from tank at 60 GPM.
- 7.5.4 Piping assembly end fittings shall be a 150 Class raised face flange per paragraph 3.14. Tank assembly shall be supplied with a blank flange closure fitting per paragraph 3.15. Temporary closure gasket for storage/shipping shall be Buna-N.

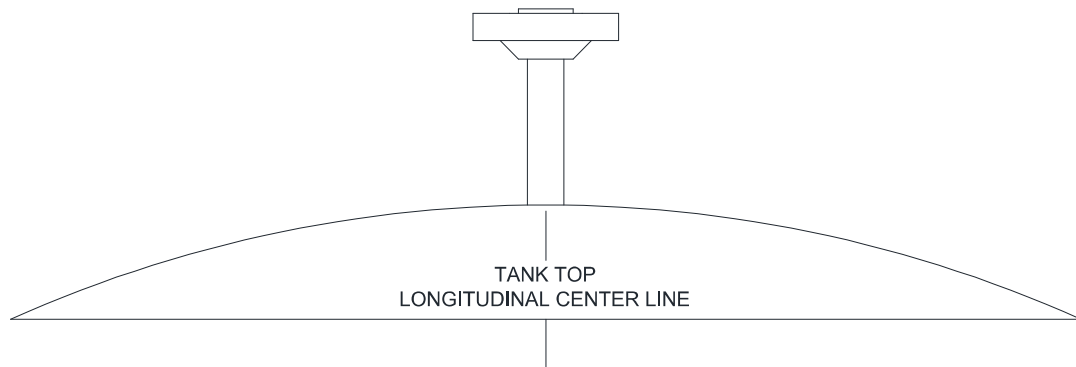
Figure 5: Nitrogen Supply and Make up Air Piping Assembly Arrangement.



- 7.6 Tank shall contain a single piping assembly connected via tank top connection/penetration for installation/connection of a relief valve assembly. See figure 6 for piping assembly arrangement.
- 7.6.1 Piping assembly shall be 1 inch SCH 40 per paragraph 3.16.
- 7.6.2 Piping assembly end fitting shall be a 150 Class raised face flange per paragraph 3.14. Piping assembly shall be supplied with a blank flange closure fitting per paragraph 3.15. Temporary closure gasket for storage/shipping shall be Buna-N.

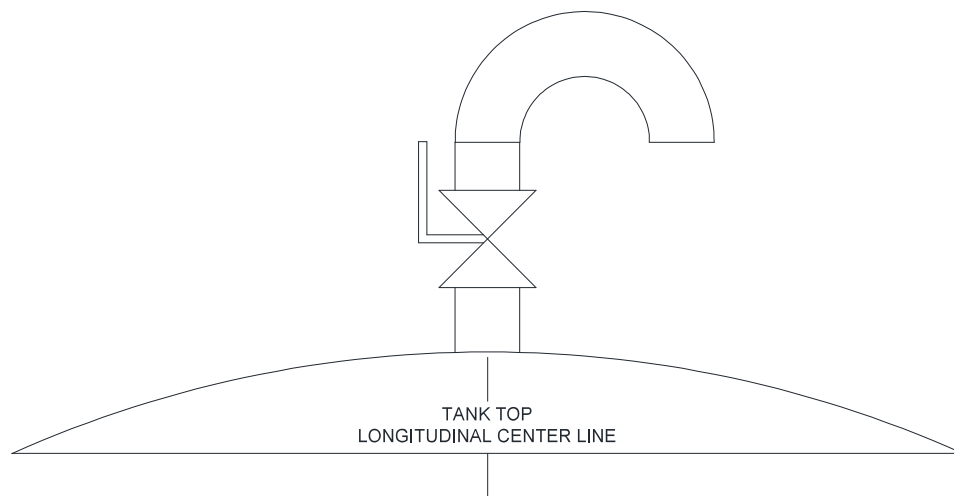
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Figure 6: Relief Valve Piping Assembly Arrangement.



- 7.7 Tank shall contain a single piping assembly connected via tank top penetration for venting the tank to atmospheric pressure supporting, filling, draining and pump down operations.
- 7.7.1 Vent assembly shall be located at the tank end opposite the fill assembly of paragraph 7.3.
- 7.7.2 Vent assembly shall be 2 inch SCH 40 per paragraph 3.16.
- 7.7.3 Vent assembly piping end open to atmosphere shall terminate in a downward facing direction (see figure 7).
- 7.7.4 A stainless steel (Grade 316) 75 micron screen shall be welded to the vent piping end open to atmosphere (alternate methods to secure mesh screen in place may be proposed in contractors drawing submitted for NNSY approval per paragraph 3.3.).
- 7.7.5 Vent assembly shall have a 2 inch full port ball valve, per paragraph 3.18, between tank penetration/connection and the piping end open to atmosphere.

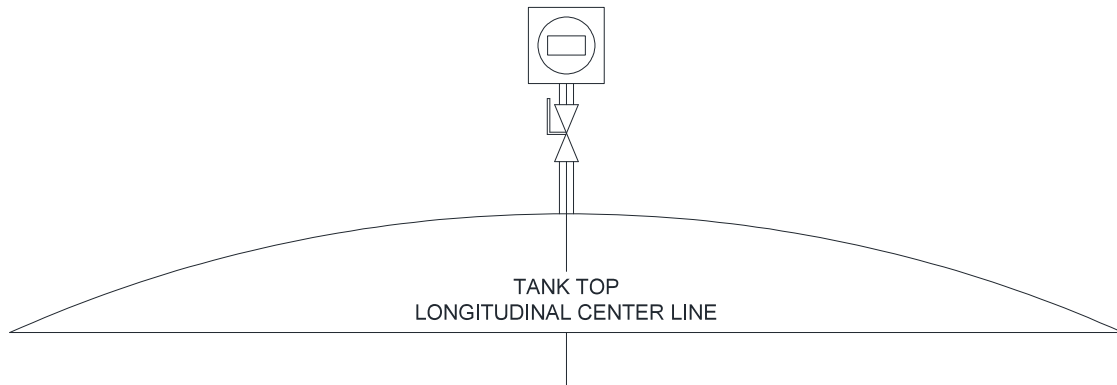
Figure 7: Tank Vent Piping Assembly Arrangement.



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- 7.8 Tank shall contain a single piping assembly connected via tank top penetration for monitoring tank internal pressure. Piping assembly shall be outfitted with a compound gauge assembly (see figure 8).
- 7.8.1 Piping assembly shall be 3/8 inch SCH 40 per paragraph 3.16.
- 7.8.2 Piping assembly shall have a 3/8 inch full port ball valve, per paragraph 3.18 between tank penetration/connection and the gauge assembly.
- 7.8.3 Gage assembly shall be a compound type (e.g. capable of indicating vacuum/suction pressure) with a digital pressure display (Ametek/Crystal Engineering XP2I 0-100 psi; VPN 100PSIXP2I or equivalent).

Figure 8: Tank Pressure Gage/Monitoring Piping Assembly Arrangement.



8 Monitoring Equipment.

- 8.1 Tank shall be outfitted with a Temposonics/MTS Level Plus (MR Model) or RefineME (LPR Model) analog liquid level/volume sensor/transmitter with resistance temperature detector located 16 inches from bottom of sensor assembly. Assembly shall utilize a rigid stainless steel probe with welded flange mounting and a 3/4" NPT conduit/cabling connection. All wetted parts of the sensing element shall be grade 304/316 stainless steel.
- 8.2 Liquid level float shall be a Temposonics/MTS float (Model 201248-2, or equivalent), constructed from grade 316/304 stainless steel having a magnet located at the top of the float to allow level indication at lowest level possible.
- 8.3 Tank volume/liquid level process meter shall be a Precision Digital (Model PD6000-6R5 with red display).
- 8.4 Tank effluent temperature process meter shall be a Precision Digital (Model PD6000-6R0 with red display). Temperature range for RTD is 20 °Fahrenheit - 240° Fahrenheit.

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- 8.5 Process meters shall be contained/mounted within a Hoffman (Model CSD242410W) concept enclosure with a NEMA 4 rating. Enclosure shall have include a concept panel (Hoffman CP2424) and a concept swing-out panel (CSP2424). Enclosure shall contain the process meters of paragraphs 8.3 and 8.4. Enclosure shall be mounted to the tank assembly via a separate free standing frame integral with the base plate our it can be mounted to the elliptical end cap where the discharge and sample connections are located. Alternate NEMA 4 rated enclosures are acceptable with NNSY approval of drawing per paragraph 3.3.
- 8.6 Main power to process meters, liquid level sensor and temperature detector shall be via a disconnect safety switch (120V AC, NEMA 4X enclosure) with associated power cord (8 ft long) outfitted with a twist lock plug end.

9 Welding Details.

- 9.1 Welding (tank shell and end caps, tank piping penetrations and any welded attachments to end caps/shell) shall be in accordance with the requirements of reference 2.1.
- 9.2 Longitudinal and circumferential welds to assemble the tank shell and circumferential joints between the shell and elliptical end caps shall be full penetration welds, welded from the outside and inside. Welds shall be back gouged to sound metal, and all slag removed prior to welding from second side.
- 9.3 Weld surfaces on interior of tank shell (including exterior welds on any piping internal to the tank) shall be ground flush and be free of weld spatter, scratches, burrs, rust and slag. Surface shall not snag or tear a cleaning cloth when wiped in any direction. Edges of welds shall be smoothly blended with base metal.
- 9.4 Welds (tank shell, end caps and any welded attachments to end caps/shell) shall be liquid penetrant inspected, at a minimum, per requirements of reference 2.1. Certification report required.
- 9.5 All structural welding shall be in accordance with references 2.20, 2.21 and 2.22 as appropriate.
- 9.6 All piping/pipe fitting welds shall be full penetration butt weld joints (i.e., use of consumable inserts allowed), per reference 2.23 or equivalent standard/specification.
- 9.7 All pipe to tank shell welds shall be full penetration welds.
- 9.8 All burrs and weld spatter shall be removed from all exterior tank and frame welds. All rough edges shall be rounded or removed.
- 9.9 Vendor shall certify that any/all weld spatter and slag has been removed and that the tank assembly internal and external surfaces a free of weld spatter and slag. No weld spatter or slag shall be present on tank assembly internal or external surfaces.

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10 Tank Assembly Testing Details.

- 10.1 Vendor shall hydrostatically pressure test the tank assembly to 33 +2/-0 psig (test boundary shall include all piping assemblies, valves and personnel access/manway). Test pressure shall be held for 30 minutes (no leaks, damage or permanent deformation allowed). Tank assembly shall have completed all non destructive testing and shall be secured in its saddle with base plate assembly during testing. Certified report required.
- 10.2 Vendor shall perform a calibrated fill of the tank to 90% fill volume (see paragraph 3.22). Level data (gallons) shall be recorded for each 1 inch change in level height. Level data shall be provided to NNSY. Certified report required.
- 10.3 Vendor shall perform level checks of the tank assembly utilizing the analog liquid level/volume sensor/transmitter of paragraph 8.1 to validate accurate operation of the sensor/transmitter and associated process meter of paragraph 8.3. Certified report required.
- 10.4 Vendor shall perform an operational check/test to validate operation and accuracy of the resistance temperature detector of paragraph 8.1 and its associated process meter of paragraph 8.4. Certified report required.
- 10.5 After completion of hydrostatic testing and level checks, the tank and piping assembly internals shall be dried by wiping and/or by blowing dry filtered air/nitrogen through internals. Certified report required.
- 10.6 Weigh entire tank assembly empty and filled to 90% capacity (tank with all piping assemblies attached and secured in its saddle with base plate assembly). Load indicating device shall be calibrated within +/- 2% of the devices full scale. Certified report required.

11 Tank Assembly Marking Requirements.

- 11.1 Tank assembly shall be marked in two locations on its saddle with base plate/frame. Marking shall provide the following information as a minimum. Markings shall be made per details of paragraph 3.9.
 - 11.1.1 Purchase Order Number
 - 11.1.2 Tank Serial Number; RDPWT-1, RDPWT-2, etc.
- 11.2 Process meters shall be labeled using solid label plates (black background with white lettering). Label plates shall be made from a hard composite material (phenolic, poly, gravoply, etc.).
 - 11.2.1 Liquid level process meter shall have a label plate stating, "Liquid Level (Gallons)".
 - 11.2.2 Temperature process meter shall have a label plate stating, "Temperature (Degrees Fahrenheit)".

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11.3 Tank assembly shall have a label plate attached to shell (one location). Label plate backing shall be constructed from stainless steel/aluminum with black block style lettering. Label plate shall contain the following information.

11.3.1 Design Pressure: 30 psig

11.3.2 Design Temperature: 120 ° F

11.3.3 Hydro Test Pressure: 33 +2/-0 psig

11.3.4 Hydro Test Date: __ - __ - ____

11.3.5 Full Tank Capacity: _____ Gallons

11.3.6 Empty Weight: _____ Pounds

11.3.7 Full Weight: _____ Pounds

11.4 Lifting attachment points shall be marked per paragraph 3.9 with the following verbiage; "LIFT POINT".

12 Vendor shall provide the following certification of compliance, inspection/test reports and additional information.

12.1 Certification of Compliance.

12.2 Inspection/Test Reports.

12.2.1 Mercury exclusion certification (see paragraph 3.5).

12.2.2 Drop Weight Test (ASTM E208) or a Charpy Impact test (ASTM A370) results/reports for integral lift lugs/rigging attachment points (see paragraph 5.3).

12.2.3 Pre and post load test magnetic particle or liquid penetrant inspection reports (see paragraph 5.6.2).

12.2.4 Lift lug/rigging attachment load test report (see paragraph 5.6.3).

12.2.5 Liquid penetrant inspection reports for tank welds (see paragraph 9.4).

12.2.6 Certify that no weld spatter or slag is present on tank internal/external surfaces (see paragraph 9.9).

12.2.7 Hydrostatic test of tank assembly (see paragraph 10.1).

12.2.8 Calibrated fill of tank to 90% fill volume (see paragraph 10.2).

12.2.9 Tank level check using analog liquid level/volume sensor/transmitter and process meter (see paragraph 10.3).

12.2.10 Operational check of resistance temperature detector and process meter (see paragraph 10.4).

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- 12.2.11 Certify tank internals fully dried after testing (see paragraph 10.5).
- 12.2.12 Certify weight of entire tank assembly empty and filled to 90% capacity (see paragraph 10.6).
- 12.2.13 All fluid contacting components/materials meet detrimental material and cleanliness requirements of references 2.2 and 2.3 (see paragraph 3.4).
- 12.2.14 All lubricants used on fluid contacting components shall meet detrimental material requirements of reference 2.2. Any factory lubricants that do not meet detrimental material requirements of reference 2.2 have been removed, surfaces cleaned per requirements of reference 2.3 and lubricants replaced with one which is compliant with reference 2.2 (see paragraph 3.4).
- 12.2.15 Certify tank shell and elliptical end cap base materials and weld areas are passivated (see paragraph 6.8).
- 12.3 Additional information.
 - 12.3.1 Operating and maintenance manual for liquid level sensor/transmitter, resistance temperature detector and process meters.
 - 12.3.2 Rigging diagram for lifting tank assembly using and overhead crane (see paragraph 5.7).
 - 12.3.3 Tank assembly drawing with material list (see paragraph 3.3).
 - 12.3.4 List of lubricants (brand name information) utilized for final assembly (i.e., inside ball valves, etc.) along with associated Safety Data Sheets (SDS) and material specifications.